

CHADA for Electron Backscatter Diffraction (EBSD)

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Abstract	Structured documentation of CHAracterization DAta (CHADA) gained by carrying out Electron Backscatter Diffraction experiments within the AddMorePower project. The structure is based on the final draft of the CWA 17815 created within the framework of the CEN/CENELEC Workshop “Materials characterization — Terminology and structured documentation”. The draft was submitted on Dec. 16, 2024 to CEN/CENELEC.
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Editor

Olivia Pfeiler (KAI)

Contributors (ordered according to beneficiary numbers)

Vedad Babic, KAI- Kompetenzzentrum Automobil und Industrieelektronik

Diana Farthofer, KAI- Kompetenzzentrum Automobil und Industrieelektronik

Sebastian Moser, KAI- Kompetenzzentrum Automobil und Industrieelektronik

Manuel Petersmann, KAI- Kompetenzzentrum Automobil und Industrieelektronik

Michael Reisinger, KAI- Kompetenzzentrum Automobil und Industrieelektronik

based on

Final Draft CWA 17815 from CEN/CENELEC WS Materials characterization — Terminology and structured documentation, submitted to CEN/CENELEC on Dec. 16, 2024.

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Executive Summary

The AddMorePower consortium is dedicated to developing characterization and modelling methods that can meet the unique needs of upcoming power semiconductor technology generations. To achieve effective and efficient research and innovation, a seamless interoperation between materials characterization, materials modelling, and data science is required. This goal can only be accomplished through the use of FAIR and open data practices.

AddMorePower employs the CHADA and MODA methodologies to achieve FAIR data management. These systematic description and documentation methods are used for materials characterization data and materials modelling data, respectively. They are based on a common terminology, concepts and relationships defined by the community, providing a holistic approach to combine data from various materials characterization and modelling techniques, which allows analyzing and modelling of complex structures.

In this document a CHADA for Electron Backscatter Diffraction (EBSD) experiments is presented, based on the final draft of the CWA 17815 created within the framework of the CEN/CENELEC Workshop “Materials characterization — Terminology and structured documentation”. The draft was submitted on Dec. 16, 2024 to CEN/CENELEC.

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Chapter 1 Introduction

The manufacturing industry is facing new challenges with the advent of Industry 5.0. Innovative solutions are required to meet these challenges. The AddMorePower consortiums focuses its research on developing necessary characterization and modelling techniques to meet the particular needs of upcoming power semiconductor technology generations. One backbone to foster research and innovation in this field is the integration of materials characterization, materials modelling and data science. To enable the fluent interoperation, FAIR and open data practices are needed.

FAIR data management requires harmonized data treatment and harmonized data documentation, not only within the facilities where the simulations and experiments are performed, but across communities. The [EMCC](#) and the [EMMC](#) developed two methodologies for a systematic description and documentation of materials CHAracterization Data (CHADA) and materials MOdelling Data (MODA). In AddMorePower we make use of the developed concepts and terminology from CHADA and MODA in order to describe the produced data in a harmonized way and to share the gained knowledge in the community.

CHADA (material CHAracterization DAta) is a systematic description and documentation method for material characterization experiments, including user case, raw data generation and post-processing of the data. It enables the systematic investigation of materials. A harmonized experiment documentation is essential for managing material science data and for guiding the development of e.g. new materials that meet specific requirements. CHADA provides the basis for a holistic approach to combine data from different characterization techniques, which enables the analysis and modelling of complex structures and links between different experiments. It is based on a common terminology, concepts and relationships defined by the community [1]. In 2024 the common terminology and structured documentation was revised by a consortium within the CEN/CENELEC workshop “Materials characterization — Terminology and structured documentation”, resulting in the final draft CWA 17815, which was submitted to CEN/CENELC on Dec. 16, 2024.

The CHADA terminology and structured documentation presented in this document is based on the final draft of the CWA 17815 from Dec. 2024. This CHADA is called “CHADA v2”

Chapter 2 What is EBSD?

Electron Backscatter Diffraction (EBSD) is a powerful technique used in conjunction with Scanning Electron Microscopy (SEM) to unravel the crystallographic structure of materials. This method involves the use of a specialized EBSD detector, which is typically composed of a phosphorescent screen, a compact lens, and a low-light camera, all integrated within a scanning electron microscope.

During an EBSD analysis, a focused beam of electrons is directed at a sample that has been carefully tilted to optimize the interaction between the electrons and the material's atoms. As the electrons collide with the sample, they are scattered in various directions, resulting in the formation of a complex pattern of backscattered electrons. These electrons, having undergone elastic diffraction and energy loss, exit the sample at distinct scattering angles and eventually reach the phosphor screen, where they form characteristic Kikuchi patterns, also known as Electron Backscatter Patterns (EBSPs).

The spatial resolution of EBSD is influenced by a multitude of factors, including the inherent properties of the material being studied, such as its crystal structure and grain size, as well as the quality of the sample preparation. The careful preparation of the sample surface and the precise control of the tilt angle are crucial in ensuring optimal EBSD results.

By analyzing the EBSPs, researchers can gain valuable insights into the material's grain structure, grain orientation, and phase composition at the micro-scale. This information is essential for understanding the material's properties and behavior.

Chapter 3 Terms and definitions from CHADA v2

For the purposes of this document, the following terms and definitions apply.

ISO and IEC maintain terminological databases for use in standardization at the following addresses:

— ISO Online browsing platform: available at <http://www.iso.org/obp/>

— IEC Electropedia: available at <http://www.electropedia.org/>

3.1 Batch

Specific quantity of material produced or processed at one time, under the same conditions.

Note 1 to entry The term batch is often used in manufacturing and production industries to track, control, and ensure the quality of materials. Each batch typically has a unique identifier and may be subject to quality control measures to ensure consistency and compliance with quality standards.

3.2 Calibration data

Output data of a measurement on a reference sample by a specific measuring instrument/device/system.

Note 1 to entry Calibration data are post-processed into corrections to be used to correct measured characterization data or to perform uncertainty calculations.

3.3 Calibration process

Operations/actions under specified conditions that are needed to obtain the correction used to convert the signal of the characterization measurement (as produced by the detector) into the raw characterization data.

EXAMPLE 1 In nanoindentation, the electrical signal coming from capacitive displacement gauge is converted into a calibrated raw-displacement data.

EXAMPLE 2 In nanoindentation, the electrical signal coming from capacitive displacement gauge is converted into a real raw-displacement signal after using a proper calibration function (as obtained by the equipment manufacturer). Then, additional calibration procedures are applied to define the point of initial contact and to correct for instrument compliance, thermal drift, and indenter area function to obtain the real usable displacement data.

Note 1 to entry Within this workflow, a possible way is to use a reference sample (with pre-defined, specific, and stable physical characteristics and known properties) for the interaction with the measurement probe, in order to extract characterization data to be compared to predefined values. In this way, the accuracy of the measurement tool and its components (for example the probe) is evaluated and confirmed.

3.4 Materials CHAracterization DAta documentation (CHADA)

Set of forms that document materials characterization.

3.5 Characterization data filtering

In data processing, process on a dataset to exclude, rearrange, or apportion data according to certain criteria.

Note 1 to entry Preprocessing of raw data (for example, compacting data, removing outliers, filtering noise, interpolating missing data) is one type of characterization data filtering.

Note 2 to entry The processing can include hardware filters (such as low pass filters "hardwired" to the sensor/probe).

3.6 Characterization data post-processing

Data analysis and transformation that allows to calculate the material property/behavior from the calibrated primary data.

Note 1 to entry Characterization data post-processing involves the application of a method, based on some theory or model, to primary data to calculate the secondary data that provide information about the characterization property or behavior.

Note 2 to entry Characterization data post-processing includes digital image processing to enhance or extract useful information.

EXAMPLE 1 In nanoindentation testing, the Oliver-Pharr method is used, which allows calculating the elastic modulus and hardness of the sample by using the load and depth measured signals.

EXAMPLE 2 Analysis of Scanning Electron Microscopy (SEM) (or optical) images to gain additional information, for example microstructural analysis, grain size evaluation, digital image correlation procedures.

3.7 Characterization data processing through calibration

Process that describes how signal data are corrected and/or modified based on calibrations into raw data.

3.8 Characterization experiment

Execution of one specific characterization workflow.

Note 1 to entry The term "test case" is typically used for a characterization experiment carried out to assess quality or performance against a defined reference or standard.

3.9 Characterization instrument

Device used for executing characterization actions to characterize a material, alone or in conjunction with one or more supplementary devices.

3.10 Characterization laboratory

Laboratory where a whole process or some of its stages take place.

3.11 Characterization measurement process

Process of experimentally obtaining data (values) that are attributed to a sample.

3.12 Characterization outcome

Digestion and/or judgement of the output of one or several tests given as feedback to the owner of the user story.

3.13 Characterization primary data

Data resulting from applying corrections to normalize and/or harmonize characterization raw data, in order to prepare them for post-processing.

3.14 Characterization raw data

Output data of the characterization measurement given by the measuring instrument.

3.15 Characterization secondary data

Data resulting from the application of characterization data post-processing.

Note 1 to entry Characterization secondary data may represent a property or a behavior.

Note 2 to entry Characterization secondary data may be used to establish a materials relation for a simulation model.

3.16 Characterization signal

Result (effect) of the interaction between the sample and the probe or between the samples, which usually is a measurable and quantifiable quantity.

Note 1 to entry Characterization signal is usually emitted from a characteristic “emission” volume, which can be different from the sample/probe “interaction” volume and can be usually quantified using proper physics equations and/or modelling of the interaction mechanisms.

Note 2 to entry According to IUPAC Compendium of Chemical Terminology, a “signal” is “A representation of a quantity within an analytical instrument”.

3.17 Characterization workflow

process composed of all necessary steps (for example, sample preparation, calibration, measurement, post-processing) to establish the property, behavior or image of a material or materials system.

3.18 Data pre-processing

process of curation (or preparing) data, which may come from disparate data sources, to improve their quality or reduce bias in subsequent analysis.

Note 1 to entry If this process is applied to characterization the input is raw data and the output is primary data.

3.19 Data processing

computation that provides a data output following the elaboration of some input data, using a data processing application possibly exploiting a data processing model.

3.20 Detector

physical device (or chain of devices) that is used to measure, quantify and store the signal generated by interaction of the probe with the sample.

EXAMPLE 1 Back Scattered Electrons (BSE) and Secondary Electrons (SE) detectors for SEM.

EXAMPLE 2 Displacement and force sensors for mechanical testing.

3.21 Environment

Surrounding medium.

EXAMPLE 1 Environment of the material or materials system that appear(s) in the user story.

EXAMPLE 2 Medium that surrounds the characterization sample in the experiment.

3.22 Hazard

set of inherent properties of a substance, mixture of substances, or a process involving substances that, under production, usage, or disposal conditions, make it capable of causing adverse effects to organisms or the environment, depending on the degree of exposure (that is, a source of danger).

3.23 Interaction volume

The volume of material, and the surrounding environment, that interact with the probe or samples and generate a detectable (measurable) signal (information).

Note 1 to entry In some cases, the volume of interaction could be different from the volume of detectable signal emission. For example, in SEM, the volume of interaction between the electron probe and the material is different from the volumes that generate the captured signal.

EXAMPLE 1 In x-ray diffraction, the interaction volume is the volume of material that interacts directly with the x-ray beam and is usually smaller than the volume of the entire sample. Depending on sample's structure and microstructure, the interaction between the sample and the x-ray incident beam generates a secondary (reflected) beam that is measured by a detector and contains information on certain sample's properties (for example, crystallographic structure, phase composition, grain size, residual stress).

EXAMPLE 2 In SEM, the interaction volume is the volume of material that directly interacts with the incident electron beam, it is usually much smaller than the entire specimen's volume and can be computed by using proper models. The interaction between the scanning probe and the sample generates a series of detectable signals (for example back scattered electrons, secondary electrons, x-rays, specimen current) which contain information on, for example, sample morphology, microstructure, and composition.

3.24 Level of automation

Degree to which tasks are executed by machines using a specified protocol.

3.25 Level of expertise

Specific experience, education, training and/or certifications required to carry out a task.

Note 1 to entry This field is typically relevant only for materials testing in regulated environments.

Note 1 to entry The level of expertise in the CHADA may just be specified as low/medium/high or in a more de- tailed way with respect, for example, to operator qualifications or certifications for certain test techniques.

3.26 materials system

Two or more materials that are in contact and/or interacting.

EXAMPLE Tribological properties (such as friction and wear) are not properties of a material, but properties of a system of two or more interacting bodies (for example, samples or components interacting with test probes or interfacial surfaces, non-destructive testing of machine components, inspection of materials in manufacturing processes). Consequently, multiple materials might be involved (for example, a polymeric and a metallic sample moving in a lubricated contact, or a sample subjected to chemical treatment).

3.27 Measurement parameters adjustment

Set of operations carried out on a characterization measurement system using a sample which is not a reference sample so that it provides values considered to be correct for a quantity being measured.

Note 1 to entry Adjustment of a measuring system is different to calibration, which is sometimes a prerequisite for adjustment. Adjusting/tuning a measuring instrument is an activity without a reference sample (which would be a calibration). The output of this process can be a specific measurement parameter to be used in the characterization measurement process.

Note 2 to entry After an adjustment of a measuring system, the measuring system is usually recalibrated.

3.28 Measurement uncertainty calculation

In data processing, the determination of the statistical dispersion of the values attributed to a measured quantity.

3.29 Operator

The human person who takes care of the entire process or any of its sub-processes/stages, including operating the test equipment.

3.30 Probe

A physical device or radiation (matter or field, together called a physical) that interacts with the sample and generates a response called signal.

Note 1 to entry	The probe is used to acquire information on the specimen's behavior and properties.
Note 2 to entry	In tribological characterization and other techniques that investigate systems, a probe may be hard to define.
EXAMPLE 1	In x-ray diffraction, the probe is a beam of x-rays with known energy that is properly focused on the sample's surface with a well-defined geometry.
EXAMPLE 2	In electron microscopy (SEM or TEM), the probe is a beam of electrons with known energy that is focused (and scanned) on the sample's surface with a well-defined beam-size and scanning algorithm.
EXAMPLE 3	In mechanical testing, the probe is a force actuator that is designed to apply a force over-time on a sample. Many variants can be defined depending on way the force is applied (for example, tensile/compressive uniaxial tests, bending test, indentation test) and its variation with time (for example, static tests, dynamic/cyclic tests, impact tests).
EXAMPLE 4	In dynamic light scattering, temporal fluctuations of backscattered light due to Brownian motion and flow of nanoparticles are the probe, resolved as function of pathlength in the sample. From fluctuation analysis (intensity correlations) and the wavelength of light in the medium, the (distribution of) diffusion coefficient(s) can be measured during flow. The Stokes-Einstein relation yields the particle size characteristics.
EXAMPLE 5	In spectroscopic methods, the probe is a beam of light with pre-defined energy (for example in the case of laser beam for Raman measurements) or pre-defined polarization (for example in the case of light beam for Spectroscopic Ellipsometry methods), that is properly focused on the sample's surface with a well-defined geometry (specific angle of incidence).

3.31 Raw data normalization

Data processing which involves adjusting raw data to a notionally common scale.

Note 1 to entry It involves the creation of shifted and/or scaled versions of the values to allow post-processing in a way that eliminates the effects of influences on subsequent properties extraction.

Note 2 to entry Raw data normalization is one type of preprocessing of raw data.

3.32 Reference sample

Sample that is sufficiently homogeneous and stable with reference to one or more specified property(ies), which has been established to be fit for its intended referencing use in measurement or in examination.

3.33 Sample

Portion of the material or materials system selected for characterization or testing.

Note 1 to entry The term 'sample' implies the existence of a sampling error, that is, the results obtained on the portions taken are only estimates of the property/behavior present in the parent material.

Note 2 to entry	Typically, the term is further qualified, for example, bulk sample, representative sample, primary sample, bulked sample, test sample.
Note 3 to entry	An alternate term is specimen. Sample and specimen are sometimes used interchangeably, or also with slightly different meanings.
Note 4 to entry	In fields such as composites, the term coupon is used to denote a sample.

3.34 Sample extraction

Act of extracting a portion (amount) of material from a larger quantity of material. This operation results in obtaining a sample representative of the batch with respect to the property or properties being investigated.

Note 1 to entry	The term can be used to cover either a unit of supply or a portion for analysis. The portion taken may consist of one or more sub-samples and the batch may be the population from which the sample is taken.
Note 2 to entry	This operation is meant to obtain a sample representative of the material with respect to the property or properties being investigated.
Note 3 to entry	The portion taken may consist of one or more sub-samples and the (sub-) samples may be the population sample representing the material.
Note 4 to entry	This can be applied to a batch of materials produced (population from which the sample is taken).

3.35 Sample holder

An object which supports the specimen in the correct position for the sample preparation.

EXAMPLE An example of sample holder is microtome holder.

3.36 Sample inspection

Analysis of the sample to determine information that is relevant for the subsequently applied characterization method.

EXAMPLE In the nanoindentation method, sample inspection is carried out by means of scanning electron microscopy to determine the indentation area.

3.37 Sample preparation

Sample preparation processes (for example, machining, polishing, cutting to size) are executed before actual observation and measurement.

3.38 Sample preparation parameter

Settings of devices used for sample preparation.

3.39 Test case

Characterization experiment carried out to assess quality or performance against a defined reference or standard.

3.40 User story

A high-level description of the problem, including the material, the environment in which it functions and the properties/behavior to be determined or explained.

Note 1 to entry There are no details related to the characterization process, which is to be decided on by the characterization expert.

Chapter 4 Terms and definitions from EBSD

For the purposes of this document, the following terms and definitions apply.

4.1 High Resolution Electron Backscatter Diffraction (HR - EBSD)

In conventional EBSD, the acquired Kikuchi or electron backscatter patterns (EBSP) are processed on-line by means of applying Hough transform, an image processing technique here used to derive lattice and crystal information from the pattern images. In HR-EBSD the EBSP are instead saved to disk for later processing by cross-correlation, a numerical technique in which the acquired patterns are compared to reference patterns. By this method, the accuracy of the derived data is substantially increased, while also allowing for additional information to be computed, most notably the local crystal strain.

4.2 Kernel Average Misorientation (KAM)

KAM is a technique in conventional EBSD for calculating the local misorientation at each measurement point by computing the average misorientation to its neighbors as defined by a kernel. The kernel can take any shape but is in MTEX typically evaluated on the n -th order neighbors i.e. KAM of order n is defined as the average misorientation of a pixel with its neighbors of order $\leq n$.

4.3 MTEX

Free [Matlab toolbox](#) for analyzing and modelling crystallographic textures by means of EBSD or pole figure data

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Chapter 5 CHADA v2 for EBSD

In this chapter the CHADA for EBSD experiments performed in the project AddMorePower is presented. The CHADA is based on the common terminology defined in the CEN Workshop Agreement 17815 [1] and the revised templates provided by the CEN/CENLEC Workshop from 2024.

CHADA authors:

Vedad Babic, KAI- Kompetenzzentrum Automobil und Industrieelektronik

Diana Farthofer, KAI- Kompetenzzentrum Automobil und Industrieelektronik

Sebastian Moser, KAI- Kompetenzzentrum Automobil und Industrieelektronik

Manuel Petersmann, KAI- Kompetenzzentrum Automobil und Industrieelektronik

Michael Reisinger, KAI- Kompetenzzentrum Automobil und Industrieelektronik

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5.1 CHADA Section 1: User story

5.1.1 General

The user story is the question or issue that shall be addressed. In table 1, the high-level goal and/or need should be described, possibly without mentioning specific characterization processes or techniques.

Table 1 —User story

User story	
Name	EBSD Microstructure Characterization
Description	After thermo-mechanical cycling, the microstructure of the Cu on polyheater devices needs to be characterized to assess damage and/or changes to the device.
Client/requester	KAI

5.1.2 Material /materials system

Table 2 should describe the material/materials system of interest. It can be a single material, a multi-materials system or a component made of one or more materials. Details about the specific material(s) or materials system and sample(s) characterised are specified in clause 5.4.

Table 2 —Material/materials system

Material	
Name	Metallization layers on polyheater devices Polyheater: Thermo-mechanical test chip for semiconductor material
Description	Copper metallizations integrated onto a Polyheater microelectronic chip Polyheaters are specifically designed test chips, to study the thermo-mechanical degradation of power metallization in semiconductor devices. The chip is composed of multiple layers of various materials, including aluminium (Al), copper (Cu) and silicon dioxide (SiO ₂), and in particular a resistive poly-silicon layer, which are grown on a Si substrate. These layers are structured to enable specific functionalities, such as rapid heat cycling and temperature monitoring.
Reference to standard	<i>(Add reference to standard, if any)</i>
Further documentation <i>(Link to a relevant document/web resource)</i>	T. Aichinger et al. [5] S. Moser et al. [6] A. Andrei et al. [1]

5.1.3 Environment and operating conditions

Table 3 should describe the environment and operating conditions in which the material or materials system is operating in its application, for example, pressure, temperature, humidity, or load.

Table 3 — Environment and operating conditions

Operational context	
Description	Polyheaters undergo thermal cycling with variable parameters: base temperature, max temperature, waiting time, pulse duration, in atmosphere, forming gas or vacuum. EBSD acquisition is performed in vacuum in scanning electron microscope.
Further documentation	S. Moser et al. [6]

5.2 CHADA Section 2: Rationale

Table 4 should specify why the characterization approach was chosen. If several tests are designed (for example, if several materials are going to be tested to find the optimal one, several loads are

tested to find the breaking point) this can be documented by considering all concepts below to be arrays. The approach should be evidenced with a peer reviewed paper or article, if possible.

Table 4 —Rationale

Rationale	
Description	Thermomechanical fatigue affects the material microstructure making EBSD the perfect characterization tool, considering EBSD characterizes phases, crystal structure and orientations.
Further documentation	S. Moser et al. [6]

5.3 CHADA Section 3: characterization workflow overview

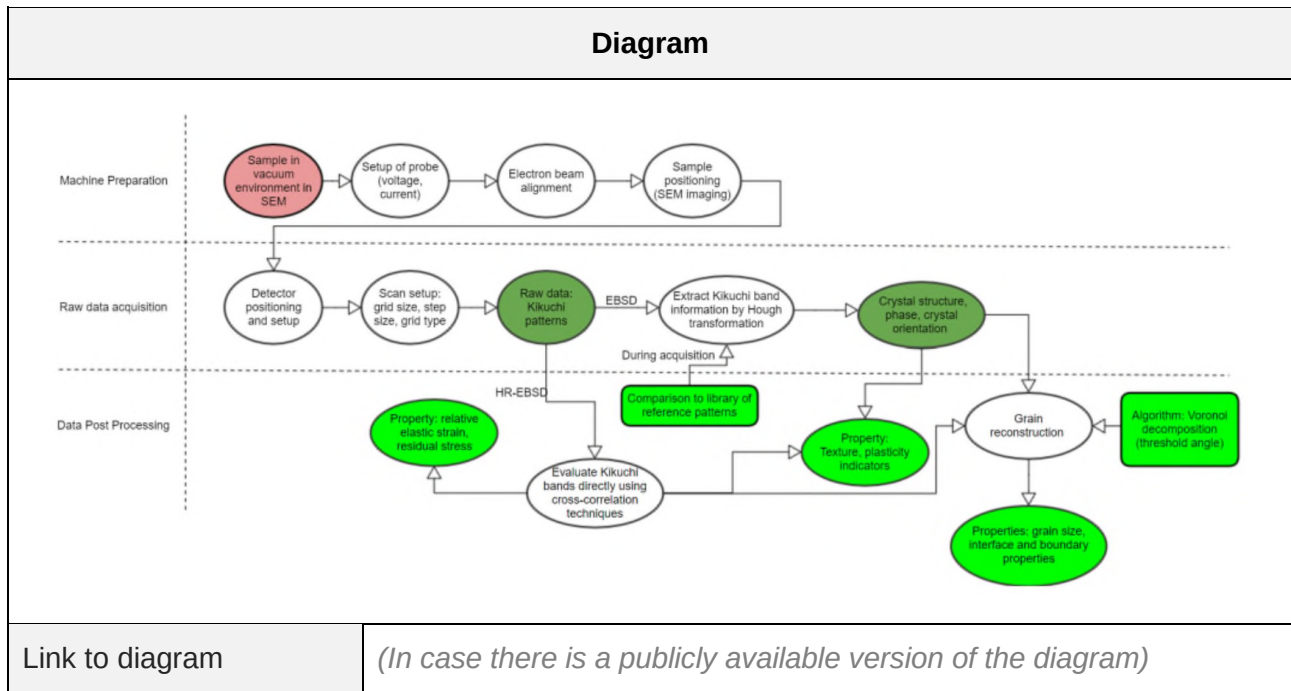
In table 5, those who employ the CHADA should provide the metadata as well as a diagram to describe the overall characterization workflow. The characterization workflow can consist of parallel experiments or tests for different materials, different external loads or time series.

The laboratory, operator, and reference to standards can be specified here if they apply to the whole procedure. If applicable, they can be also indicated for each task separately in the respective tables.

Table 5— Characterization workflow metadata

Characterization Workflow metadata	
Name	Polyheater microstructure characterization
Description	The standard workflow used in KAI for polyheater microstructure characterization using EBSD
Sample(s) to be tested	Polyheater samples, identified with Sample ID number Starting with P and then four numbers PXXXX sample XXXX (e.g.: P1111)
Test method(s) and environment(s)	see Table 6— Characterization workflow diagram
Reference to standard	E.g. ISO13067:2020 for grain size analysis using EBSD
Laboratory	KAI MSS
Operator	KAI MSS expert name
Further documentation	<i>(Link to a relevant document/web resource)</i>

Table 6— Characterization workflow diagram



5.4 CHADA Section 4: characterization experiment/ test case documentation

5.4.1 General

Each experiment and/or test procedure should be documented by using the following template tables. If several materials or samples are tested at one time or if a time series is measured, the measurements can be kept together by considering entities to be lists or arrays.

Time series measurements can be document by considering all boxes to be arrays. In this way, all measurements in a set can be documented together in one CHADA.

Table 7 and table 8 document the material/materials system and sample(s) to be investigated in some detail. These have grey headers.

In tables 9 to 16, and in the BPMN diagram, the characterization workflow is documented, consisting of different processes (shown with blue headers) and datasets (magenta) as inputs and outputs. Here the material(s) and sample(s) are referenced just by data their unique identifier and/or unique name.

5.4.2 Material/materials system

In table 7, the material, piece, volume, batch or materials system from which the sample/sample parts is extracted.

Table 7—Material/materials system

Material/materials system	
Name	Semiconductor wafer with Polyheater devices
Identifier(s)	Lot number & wafer number
Description	Typical several wafers are produced in parallel as a single lot. The individual wafers differ within the production parameter including Cu electrolyte and Cu thickness. One wafer consists of thousands of polyheater chips, which different chip designs, enabling the testing of “nominally” identical samples with different characterization methodologies.
Process history	Wafers are produced in a wafer level manufacturing process at Infineon Technologies Austria. <i>(including manufacturing steps or other prior processing information that describes the state of the material)</i>
Physical structure	Wafer with multiple layers of various materials, including aluminium (Al), copper (Cu) and silicon dioxide (SiO ₂), which are grown on a Si substrate. The main layer of interest is Cu. Detailed physical structure is explained in Table 8
Composition / Chemical structure	Electrochemically deposited thin film polycrystalline copper metallization up to 20µm thickness. Depending on electrolyte and deposition parameters, wafers with different Cu-impurities are produced.

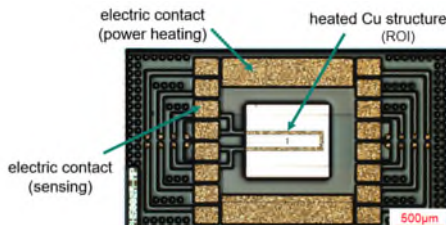
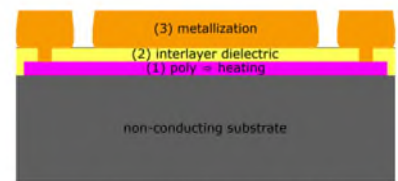
Properties known prior to the characterization	While silicon has a lower X-ray absorption coefficient than copper, it also absorbs the majority of X-rays if thicker than 100 μm . Additionally, even 5 μm of copper absorb a significant amount of light, 20 μm absorb even more and lower the transmission down to an almost negligible amount. Therefore, the thickness of the silicon substrate should be as minimal as possible while still achieving structural integrity and the copper thickness should be at the lower end of the interesting range. Check also with this X-ray database: https://henke.lbl.gov/optical_constants/filter2.html
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5.4.3 Sample

Table 8 has two fields to label the sample: name and identifier. The name is typically used in a human understandable way to uniquely label the sample status in different steps of the sample preparation, for example, raw sample, prepared (for example, polished) sample. The identifier is typically an alphanumeric code that is used as a machine-readable way to identify and trace a sample throughout the steps of preparation and hence may be the same in different steps.

Table 8— Sample

Sample	
Name	Polyheater device
Identifier	Sample ID number Starting with P and then four numbers PXXXX sample XXXX (e.g.: P1111)
Description	Polyheaters are semiconductor test chips, to enable thermo-mechanical stress testing of Cu metallization layers. The Cu metallization structure is to be investigated ($\triangle$ ROI).
Reference to standard	<i>(Add reference to standard, if any)</i>
Further documentation	A detailed description of the samples can be found in the papers from T. Aichinger et al. [5], S. Moser et al. [6], A. Andrei et al. [1]
Dimensions	The region of interest consists of polycrystalline copper which degrades under thermo-mechanical stress and voids and cracks develop. The dimension of the Cu structure within the region of interest varies for different polyheater types and wafers. Its shape (including dimensions) is listed within the corresponding polyheater design documentation, whereas the Cu thickness is documented in the wafer processing details (all dimensions in μm).
Physical structure	A single polyheater chip consists of a 30 - 120 μm Si substrate and several functional layers. A high resistive polycrystalline silicon layer (poly-Si) is electrically isolated via SiO_2 layers from the surrounding. Via Joule heating the poly-Si can heat the central area of the chip. Close by, an aluminium line structure enables temperature monitoring via

	resistance measurement. Finally, a 3 and 20 μm thick copper power metallization layer is electro-chemically deposited. This copper metallization is structured and consists of several functional pads (for contacting and actuating the chip) and a Cu structure located within the heated area. This Cu structure is the main region of interest in terms of studying microstructural changes during thermo-mechanical fatigue testing. No conductive coating needed.   Polyheater top view Polyheater cross-section
Material composition / Chemical structure	See Table 7—Material/materials system
Other sample properties	
Hazard	Don't eat sample. Follow safety procedures for electropolishing. Fragile, scratches easily, brittle

5.4.4 Sample extraction

Table 9— Sample extraction

Sample Extraction		
Name (unique)	Picking a Polyheater chip from full wafer	
Description	Individual wafers differ in production parameters such as Cu thickness, Cu type and Si substrate thickness. A single wafer contains polyheater chips with different designs, dedicated for specific experiments. During production wafers are glued on a foil and diced to separate individual chips. Hence a sample can be picked easily from the dicing foil manually.	
Input: materials data	Material	Functional polyheater chip
	Provider	Manufacturer (Infineon Technologies Austria, KAI)
	Identifier	Cu type/thickness, Polyheater design, Number of cycles, cycling conditions,
	Description	Thickness of silicon substrate, thickness of copper metallization, number of heating cycles

Laboratory	KAI MSS	
Operator	KAI MSS polyheater expert name	
Instrument	Name	None, manual step
	Manufacturer	
	Model	
	Unique ID	
Level of expertise of the operator and/ or laboratory capabilities	experienced staff	
Level of automation	manual; not automated	
Reference to standard	<i>(Add reference to standard, if any)</i>	
Further documentation	<i>(Link to a relevant document/web resource)</i>	
Output (Sample)	Sample ID (PXXXX), with info on whether specified manufacturing (thinning, cutting out) and testing conditions has been applied	

5.4.5 Sample preparation

In table 10, the “instrument” information concerns the preparation of the sample.

Table 10 — Sample preparation (1)

Sample preparation (1)		
Name (unique)	Mechanical polishing	
Description	Semi-automatic grinding/polishing system.	
Input: sample data	Polyheater sample <i>(unique name or ID of sample, see also sample section for a detailed description of the sample)</i>	
Input	<i>(from other steps of this characterization procedure, other procedures or external data)</i>	
Laboratory	KAI MSS	
Operator	KAI MSS polyheater expert name	
Instrument	Name	Allied Multi-Prep™

	Manufacturer	Allied
	Model	Multi-Prep™
	Unique ID	
Holder	<i>holder name</i>	
Parameters	<i>(settings for the preparation)</i>	
Environment	Description	
	Properties	
Level of expertise of the operator and/or laboratory capabilities	Student with special training <i>(with reference to relevant standards)</i>	
Level of automation	Manual; not automatable	
Reference to standard	See procedure and instructions at the sample preparation lab.	
Further documentation	<i>(Link to a relevant document/web resource)</i>	
Output: prepared sample data	Polyheater	

Table 11 — Sample preparation (2)

Sample preparation (2)		
Name (unique)	Electro-polishing	
Description	xxx	
Input: sample data	Polyheater sample <i>(unique name or ID of sample, see also sample section for a detailed description of the sample)</i>	
Input	<i>(from other steps of this characterization procedure, other procedures or external data)</i>	
Laboratory	KAI MSS	
Operator	KAI MSS polyheater expert name	
Instrument	Name	
	Manufacturer	KAI
	Model	Custom made

	Unique ID	
Holder	<i>Holder name</i>	
Parameters	Voltage = 2 V Polishing time: Variable (<i>settings for the preparation</i>)	
Environment	Description	
	Properties	
Level of expertise of the operator and/or laboratory capabilities	Master student with special training (<i>with reference to relevant standards</i>)	
Level of automation	Manual; not automatable	
Reference to standard	See procedure and instructions at the sample preparation lab.	
Further documentation	<i>(Link to a relevant document/web resource)</i>	
Output: prepared sample data	Polyheater	

5.4.6 Sample inspection

Sample inspection is usually carried out through a characterization technique. It is not meant to determine the investigated material's characterization properties, rather to evaluate aspects that are relevant to the characterization measurement (for example, the sample conditions and quality, the area of interest for the characterization measurement).

Table 12— Sample inspection

Sample inspection	
Name (unique)	Scanning electron microscope, SEM
Description	SEM imaging is used prior to activating the EBSD acquisition to find specific ROI
Input: sample data	Polyheater sample (<i>unique name or ID of sample, see also sample section for a detailed description of the sample</i>)
Input	(<i>from other steps of this characterization procedure, other procedures or external data</i>)
Laboratory	KAI MSS

Operator	KAI MSS polyheater expert name	
Instrument	Name	TF APREO2S
	Manufacturer	Thermofisher scientific
	Model	APREO2S
	Unique ID	
Parameters	Acceleration voltage, current, working distance	
Level of expertise of the operator and/or laboratory capabilities	Master student, with special training Should know how to operate both SEM and EBSD acquisition	
Level of automation	Not automated	
Reference to standard	<i>(Add reference to standard, if any)</i>	
Further documentation	<i>(Link to a relevant document/web resource)</i>	
Output	ROI found on the sample.	

5.4.7 Calibration

Table 13 – Calibration

Calibration		
Name (unique)	EBSD Calibration	
Description	EBSD was calibrated at installation	
Input: reference sample data	<i>(specify in case of a reference sample, reference material or a certified reference material)</i>	
Input	<i>(from other steps of this characterization procedure, other procedures or external data)</i>	
Laboratory		
Operator		
Instrument	Name	
	Manufacturer	
	Model	
	Unique ID	

Parameters	<i>(settings for the preparation)</i>
Level of expertise of the operator and/or laboratory capabilities	<i>(with reference to relevant standards)</i>
Level of automation	
Reference to standard	<i>(Add reference to standard, if any)</i>
Further documentation	<i>(Link to a relevant document/web resource)</i>
Output: calibration data	

5.4.8 Measurement parameters adjustment

Table 14 – Measurement parameters adjustment

Measurement parameters adjustment		
Name (unique)	EBSD parameter adjustment	
Description	EBSD parameter adjustment prior to acquisition. Perform spot measurement to inspect Kikuchi patterns and adjust the following parameters to achieve nice patterns: Tilt angle, tilt axis, insertion distance, detector orientation, background correction, camera binning mode, exposure time, camera gain, step size, grid size, grid type, bounding box size, phases, hit rate, site label	
Input: sample data	Polyheater sample with Sample ID <i>(unique name/id of the sample, see also sample section)</i>	
Input	<i>(from other steps of this characterization procedure, other procedures or external data)</i>	
Laboratory	KAI MSS	
Operator	KAI MSS polyheater expert name	
Instrument	Name	
	Manufacturer	
	Model	
	Unique ID	
Parameters	<i>(settings for the preparation)</i>	

Level of expertise of the operator and/or laboratory capabilities	Master student, with special training Should know how to operate both SEM and EBSD acquisition <i>(with reference to relevant standards)</i>
Level of automation	
Reference to standard	<i>(Add reference to standard, if any)</i>
Further documentation	<i>(Link to a relevant document/web resource)</i>
Output: adjusted settings for the instruments	Tilt angle, tilt axis, insertion distance, detector orientation, background correction, camera binning mode, exposure time, camera gain, step size, grid size, grid type, bounding box size, phases, hit rate, site label

5.4.9 Measurement Process

Table 15 – Measurement Process

Measurement Process		
Name (unique)	EBSD acquisition	
Description	The electron beam impinges on the sample surface and subsurface, where it interacts with the probed sample material. The backscattered electrons are detected in the EBSD-detector creating Kikuchi diffraction patterns. They are automatically analyzed to extract the crystal structure, phase, and orientation information and stored together with the SEM image of the EBSD scanned area.	
Input: sample data	Polyheater sample with Sample ID <i>(unique name/id of the sample, see also sample section)</i>	
Input	<i>(from other steps of this characterization procedure, other procedures or external data)</i>	
Laboratory	KAI MSS	
Operator	KAI MSS polyheater expert name	
Instrument	Name	EBSD detector
	Manufacturer	Oxford Instruments
	Model	C-Nano
	Unique ID	
Parameters	<i>(settings for the preparation)</i>	

Probe	Electron beam	
Interaction volume	The scattering interaction is limited to the surface and subsurface layers of the sample. The volume of interaction, in particular the depth of interaction, strongly depends on the probing parameters such as the acceleration voltage.	
Detector	Solid state detector using either a Charge Coupled Device (CCD) sensor or a Complementary Metal-Oxide-Semiconductor (CMOS) sensor for detecting diffraction patterns of back-scattered electrons.	
Environment	Description	SEM chamber
	Properties	high vacuum (1E-5 – 1E-6 mbar)
Characterization signal	Back-scattered electrons	
Level of expertise of the operator and/or laboratory capabilities	Expert SEM microscopist <i>(with reference to relevant standards)</i>	
Level of automation	Manual (could be automated)	
Reference to standard	<i>(Add reference to standard, if any)</i>	
Further documentation	<i>(Link to a relevant document/web resource)</i>	
Output: raw data	Kikuchi patterns (.h5oia)	

5.4.10 Data processing

Table 15 should be used for all types of data processing, including data normalisation, data filtering, post- processing, measurement uncertainty calculation, calibration.

A series of table 15 should be used for the respective data processing tasks specified in the “Type of data processing” entry.

Table 16 — Data processing (1)

Data processing (1)	
Name	HR-EBSD
Description	Image processing of Kikuchi patterns to get higher resolution orientation maps and strain maps.
Input	Kikuchi patterns (.h5oia) <i>(raw data, primary data)</i>
Type of data processing	Post-processing
Data processing model	

Data processing software	CrossCourt4
Laboratory	Evaluation PC at KAI MSS
Operator	KAI MSS polyheater expert name
Level of expertise of the operator and/or laboratory capabilities	Should know how to operate CrossCourt4 HR-EBSD software <i>(with reference to relevant standards)</i>
Level of automation	Manual, but could be automated
Reference to standard	<i>(Add reference to standard, if any)</i>
Further documentation	<i>(Link to a relevant document/web resource)</i>
Output	Pre-processed SEM images in .tiff format

Table 17 – Data processing (2)

Data processing (2)	
Name	EBSD post-processing
Description	Process each Kikuchi pattern (Hough transform) to assess crystal orientations
Input	Kikuchi patterns from EBSD detector
Type of data processing	Post-processing
Data processing model	
Data processing software	Aztec Crystal
Laboratory	KAIMSS – Oxford computer
Operator	KAI MSS polyheater expert name
Level of expertise of the operator and/or laboratory capabilities	Should know how to operate Aztec Crystal <i>(with reference to relevant standards)</i>
Level of automation	Automated
Reference to standard	<i>(Add reference to standard, if any)</i>
Further documentation	<i>(Link to a relevant document/web resource)</i>
Output	Processed EBSD data (.h5oia) - crystal structure, orientation, mineral phase

Table 18 – Data processing (3)

Data processing (3)	
Name	Grain reconstruction and data clean-up
Description	Using orientation and phase data acquired from the EBSD, a grain reconstruction algorithm, based on Voronoi decomposition, is performed twice, first to get prior grain size information, after which small grains, typically <3 px in size are deleted and follow a secondary grain-reconstruction is performed to get representative grain size information. Interpolation of orientation and phase data for deleted pixels from neighboring data points. Optional: Orientation filtering to reduce measurement noise
Input	Acquired EBSD data from detector (.h5oina)
Type of data processing	Data pre-processing, filtering
Data processing model	
Data processing software	MTEX
Laboratory	Personal computer at KAI
Operator	KAI MSS polyheater expert name
Level of expertise of the operator and/or laboratory capabilities	Should know how to operate EBSD processing software <i>(with reference to relevant standards)</i>
Level of automation	Semi-automated
Reference to standard	ISO 13067:2020, EBSD for grain size analysis
Further documentation	<i>(Link to a relevant document/web resource)</i>
Output	Processed EBSD maps + grains data (.h5inf)

5.5 CHADA section 5: results analysis

Table 18 should be used to show the digestion of the output (post-processed data) of all relevant characterization procedures to get to the material properties and/or behaviour that were requested in the user story in clause 5.1. It should include the interpretation and explanation of results.

Characterization results		
	Input	Processed EBSD maps + grains data (.h5inf)

Characterization results 1	Method	Compute the grain sizes via area weighted or arithmetic method (Mean +/- standard deviation) <i>(description including method by which results were obtained across several characterization procedures or test series)</i>
	Output	Grain size mean +/- standard deviation, list of grain sizes
	Further documentation	<i>(Link to a relevant document/web resource)</i>
Characterization results 2	Input	Processed EBSD maps + grains data (.h5inf)
	Method	Intragranular plastic deformation can be assessed by local misorientation analysis from processed EBSD data
	Output	Plastic deformation
Characterization results 3	Input	Processed EBSD maps(.h5oia)
	Method	Evaluating the distribution of crystal orientations on the sample to extract information about the texture
	Output	Texture evaluation
	Further documentation	<i>(Link to a relevant document/web resource))</i>

Chapter 6 List of Abbreviations

Abbreviation	Translation
AlGaN	Aluminium Gallium Nitride
ATEX	Analysis Tools for Electron and X-ray diffraction
CCD	Charge Couples Device
CHADA	Materials Characterization Data
CMOS	Complementary Metal-Oxide-Semiconductor
EBSD	Electron Backscatter Diffraction
EMCC	European Materials Characterization Council
EMMC	European Materials Modelling council
FWBW	Forward backward difference scheme
GaN	Gallium Nitride
HR-EBSD	High Resolution Electron Backscatter Diffraction
IPF	Inverse Pole Figure
KAI	Kompetenzzentrum Automobil- und Industrieelektronik
KAI MSS	KAI Material Science & Simulation team
KAM	Kernel Average Misorientation
kV	kilovolt
MAD	Mean Angular Deviation
ms	milliseconds
MTEX	Free Matlab toolbox for analyzing and modeling crystallographic textures by means of EBSD or pole figure data
RMS	Root Mean Square
ROI	Region of Interest
SEM	Scanning Electron Microscopy
Si	Silicon
ULM	Université de Lorraine

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